

Software Defined Networking (SDN) Platforms: The Efficient Eight



Software defined networking (SDN) is an emerging technology in the field of cloud computing. It is highly dynamic, manageable, cost-effective and adaptable, and is well suited to high-bandwidth, dynamic applications. Many SDN platforms are available—both open source and commercial. Selecting a platform suitable to the requirements of the customer is a tedious task. This article gives an overview of eight open source SDN platforms, so that informed decisions can be made.

Since 2016, software defined networking (SDN) has continued to evolve rapidly and become more mature, redefining corporate networking. Recently, IDC published a study which shows that the SDN market could be valued at around US\$ 12.5 billion by 2020. Cloud computing, in addition to third party platforms, has increased the need for SDN, and this need will rise further considering the increase in network-virtualisation software, and SDN applications like virtualised networks and security services.

SDN is becoming popular because networks now lag behind other areas of infrastructure, especially in the computing and storage arenas. Modern advancements like virtualisation, distributed architectures, Big Data and cloud computing require networking that can adapt and be

optimised on-the-fly using centralised intelligence. Various commercial and open source SDN platforms are available, so selecting a platform as per the requirements of the customer is a tedious task. This article aims to simplify the selection task.

Introducing software defined networking

The Open Networking Foundation (ONF) is a non-profit consortium dedicated to designing, developing, standardising and commercialising SDN. According to ONF, "Software defined networking (SDN) is an emerging network architecture where network control is decoupled from forwarding and is directly programmable."

SDN basically refers to the ability of software applications to program individual network devices